



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fourth Semester B.E. (E&E) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

20EE404 – ELECTROMAGNETIC FIELDS

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **ONE** questions from each Unit.

2) Use of Formula Table is Permitted.

Unit – I

	Marks	BT*	CO*	PO*
1. a) Explain the experimental law of coulomb in vector form considering two-point charges.	6	L*1	1	1,2
b) Calculate the total charge within each of the indicated volume $0.1 \leq x , y , z \leq 0.2; \rho_v = \frac{1}{x^3 y^3 z^3}$	6	L2	1	1,2
c) Deduce the expression for Electric field intensity at any point due to four point charges.	8	L2	1	1,2
2. a) State and Prove Gauss's Law.	8	L1	1	1,2
b) A charge of $-0.3\mu C$ is located at $A(25, -30, 15)$ (in cm), and a second charge of $0.5\mu C$ is at $B(-10, 8, 12)$ cm. Find $\vec{E}$ at: (i) the origin; (ii) $P(15, 20, 50)$ cm.	8	L2	1	1,2
c) State and Explain Divergence Theorem.	4	L1	1	1,2

Unit – II

3. a) Derive and explain about the energy expended in moving a point charge in an Electric Field.	6	L2	2	1,2
b) Given the electric field $\vec{E} = \frac{1}{z^2} (8xyz\vec{a}_x + 4x^2z\vec{a}_y - 4x^2y\vec{a}_z) V/m$ Find the differential amount of work done in moving a $6nC$ charge a distance of $2\mu m$, starting at $P(2, -2, 3)$ and proceeding in the direction $\vec{a}_L = \frac{3}{7}\vec{a}_x + \frac{6}{7}\vec{a}_y$.	6	L2	2	1,2
c) Define Potential and Potential Difference using supporting Equations.	8	L1	2	1,2
4. a) Explain the concept of continuity of current. Demonstrate the same by selecting $t = 1s, r = 5m$ and $r=6m$.	6	L2	2	1,2
b) Current density is given in cylindrical coordinates as $\vec{J} = -10^6 z^{1.5} \vec{a}_z A/m^2$ in the region $0 \leq \rho \leq 20\mu m$; for $\rho \geq 20\mu m, \vec{J} = 0$. (i) Find the total current crossing the surface $z = 0.1m$ in the $\vec{a}_z$ direction (ii) If the charge velocity is $2 \times 10^6 m/s$ at $z = 0.1m$ find ρ_v there. (iii) If the volume charge density at $z = 0.15m$ is $-2000 C/m^3$, find the charge velocity there.	8	L2	2	1,2

- c) Explain the boundary conditions for perfect dielectrics

6 L1 2 1,2

Unit – III

5. a) From fundamentals derive the Laplace's equation. Using Laplace's equation, find an expression for capacitance of a parallel plate capacitance.

10 L2 3 1,2

- b) A current filament carrying 15A in $\vec{a}_z$ direction lies along entire z-axis. Find $\vec{H}$ in rectangular coordinates at (i) $P_A(\sqrt{20}, 0, 4)$, (ii) $P_B(2, -4, 4)$

10 L2 3 1,2

6. a) Using Ampere Circuital Law deduce the expression for magnetic field intensity for coaxial cable carrying a uniformly distributed current I in the inner conductor and -I in the outer conductor. Choose a circular path of radius (i) smaller than inner conductor (ii) within the outer conductor.

8 L1+
L2 3 1,2

- b) Evaluate the closed line integral of $\vec{H}$ about the rectangular path $P_1(2, 3, 4)$ to $P_2(4, 3, 4)$ to $P_3(4, 3, 1)$ to $P_4(2, 3, 1)$ to P_1 , given $\vec{H} = 3z\vec{a}_x - 2x^3\vec{a}_z$ A/m.

Also determine $(\nabla \times \vec{H})_y$ at the center of the area.

6 L3 3 1,2

- c) State and Explain Stokes Theorem.

6 L2 3 1,2

Unit – IV

7. a) State and Explain Lorentz force equation.

6 L2 4 1,2

- b) Obtain the magnetic boundary conditions at the interface between two magnetic materials.

8 L2 4 1,2

- c) Explain concept of displacement current.

6 L1 4 1,2

8. a) List Maxwell's Equations in point and integral form.

10 L2 4 1,2

- b) Derive an expression for inductance per meter length of a coaxial of inner radius 'a' and outer radius 'b' from the basics.

10 L1+
L2 4 1,2

Unit – V

9. a) Explain lossless propagation of sinusoidal voltages.

10 L2 5 1,2

- b) Explain voltage standing wave ratio.

10 L2 5 1,2

10. a) State and Explain Poynting's Theorem and Wave Power.

10 L2 5 1,2

- b) Explain the propagation in good conductors. Elaborate Skin Effect.

10 L2 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

∇ Operator Table**Divergence**

Rectangular

$$\nabla \cdot \vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

Cylindrical

$$\nabla \cdot \vec{D} = \frac{1}{\rho} \frac{\partial(\rho D_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$

Spherical

$$\nabla \cdot \vec{D} = \frac{1}{r^2} \frac{\partial(r^2 D_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(D_\theta \sin \theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial D_\phi}{\partial \phi}$$

Gradient

Rectangular

$$\nabla V = \frac{\partial V}{\partial x} \vec{a}_x + \frac{\partial V}{\partial y} \vec{a}_y + \frac{\partial V}{\partial z} \vec{a}_z$$

Cylindrical

$$\nabla V = \frac{\partial V}{\partial \rho} \vec{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \vec{a}_\phi + \frac{\partial V}{\partial z} \vec{a}_z$$

Spherical

$$\nabla V = \frac{\partial V}{\partial r} \vec{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \vec{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \vec{a}_\phi$$

Laplacian

Rectangular

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

Cylindrical

$$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

Spherical

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \theta^2} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \phi} \left(\sin \theta \frac{\partial V}{\partial \phi} \right)$$

Curl

Rectangular

$$\nabla \times \vec{H} = \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) \vec{a}_x + \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) \vec{a}_y + \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \vec{a}_z$$

Cylindrical

$$\nabla \times \vec{H} = \left(\frac{1}{\rho} \frac{\partial H_z}{\partial \phi} - \frac{\partial H_\phi}{\partial z} \right) \vec{a}_\rho + \left(\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial \rho} \right) \vec{a}_\phi + \frac{1}{\rho} \left[\frac{\partial(\rho H_\phi)}{\partial \rho} - \frac{\partial H_\rho}{\partial \phi} \right] \vec{a}_z$$

Spherical

$$\nabla \times \vec{H} = \frac{1}{r \sin \theta} \left(\frac{\partial(H_\phi \sin \theta)}{\partial \theta} - \frac{\partial H_\theta}{\partial \phi} \right) \vec{a}_r + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial H_r}{\partial \phi} - \frac{\partial(r H_\phi)}{\partial r} \right) \vec{a}_\theta + \frac{1}{r} \left[\frac{\partial(r H_\theta)}{\partial r} - \frac{\partial H_r}{\partial \theta} \right] \vec{a}_\phi$$
